Project Report: Heart Disease Analysis

1. INTRODUCTION

1.1 Project Overview

This project focuses on analysing a heart disease dataset to identify key risk factors and patterns that influence cardiovascular health. Using Tableau for data visualization, the project transforms raw health data into meaningful insights that support better clinical decisions and preventive health strategies.

1.2 Purpose

- To understand how demographic, lifestyle, and medical attributes impact heart disease.
- To develop interactive dashboards and story points for effective data storytelling.
- To provide actionable insights for researchers, healthcare professionals, and policy makers.

2. IDEATION PHASE

2.1 Problem Statement

Heart disease remains one of the leading causes of death globally. Early detection relies on recognizing risk factors. However, traditional methods lack visual interpretation.
How can data visualization help uncover the major predictors of heart disease for better preventive strategies?

2.2 Empathy Map Canvas

- **Think & Feel:** Patients worry about symptoms and long-term health.
- **See:** Increasing heart-related cases in the community.
- **Hear:** Doctors advising lifestyle changes.
- **Say & Do:** Patients often ignore early signs.
- **Pain:** Lack of awareness about risk factors.
- **Gain:** A clear visual dashboard showing their health risks.

2.3 Brainstorming

Ideas generated:

- Visualizing demographic risks (age, gender, race)
- Mapping lifestyle impacts (smoking, alcohol, exercise)

- Identifying comorbidities (diabetes, stroke, kidney disease)
- Creating a predictor summary heatmap
- Storytelling through Tableau Story Points

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stages: Awareness → Screening → Diagnosis → Prevention
Users: General public, healthcare providers, analysts
Needs: Quick insights, easy-to-read visuals, data-backed decisions

3.2 Solution Requirements

- Clean and organized dataset
- Interactive dashboards (filters, charts)
- Story points to explain insights
- Insightful visualizations (bar, heatmap, pie chart, relationships)

3.3 Data Flow Diagram

Level 0 DFD (simple):
User → Data Input → Processing (Cleaning + Analysis) → Visualization → Story Output

3.4 Technology Stack

- **Frontend / Visualization:** Tableau
- **Backend / Processing:** Tableau Prep
- **Dataset Source:** Heart Disease dataset
- **Documentation:** MS Word

4. PROJECT DESIGN

4.1 Problem-Solution Fit

Problem: Lack of clear understanding of heart disease risk factors.
Solution: Interactive visual analytics to interpret risk easily.

4.2 Proposed Solution

- Visualize demographic, medical, and lifestyle parameters
- Build dashboards summarizing key influencers
- Use Tableau Story for sequential storytelling

- Provide insights to reduce risk

4.3 Solution Architecture

Data Source → Preprocessing → Tableau Worksheets → Dashboards → Story Points → Final Report

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phases:

1. Requirement Gathering
2. Data Cleaning
3. Visualization Creation
4. Dashboard Development
5. Story Design
6. Report Documentation

Timeline: 2 weeks

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

- Checked dashboard load speed
- Verified filter responsiveness
- Ensured story navigation smoothness
- Validated correctness of numbers (counts, totals, groupings)
- Tested across different screen sizes

7. RESULTS

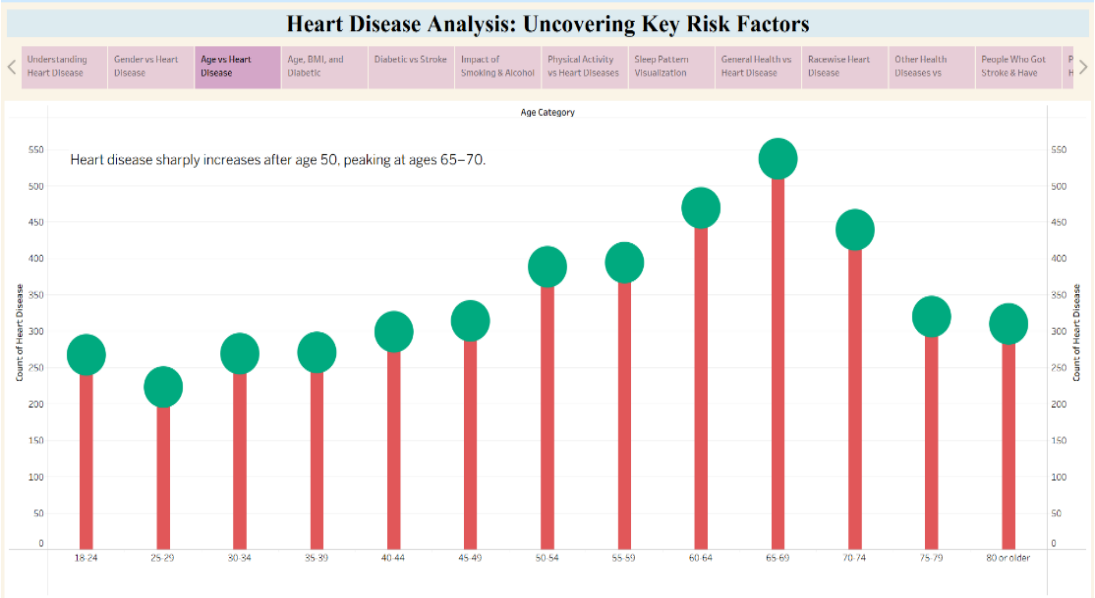
7.1 Output Screenshots

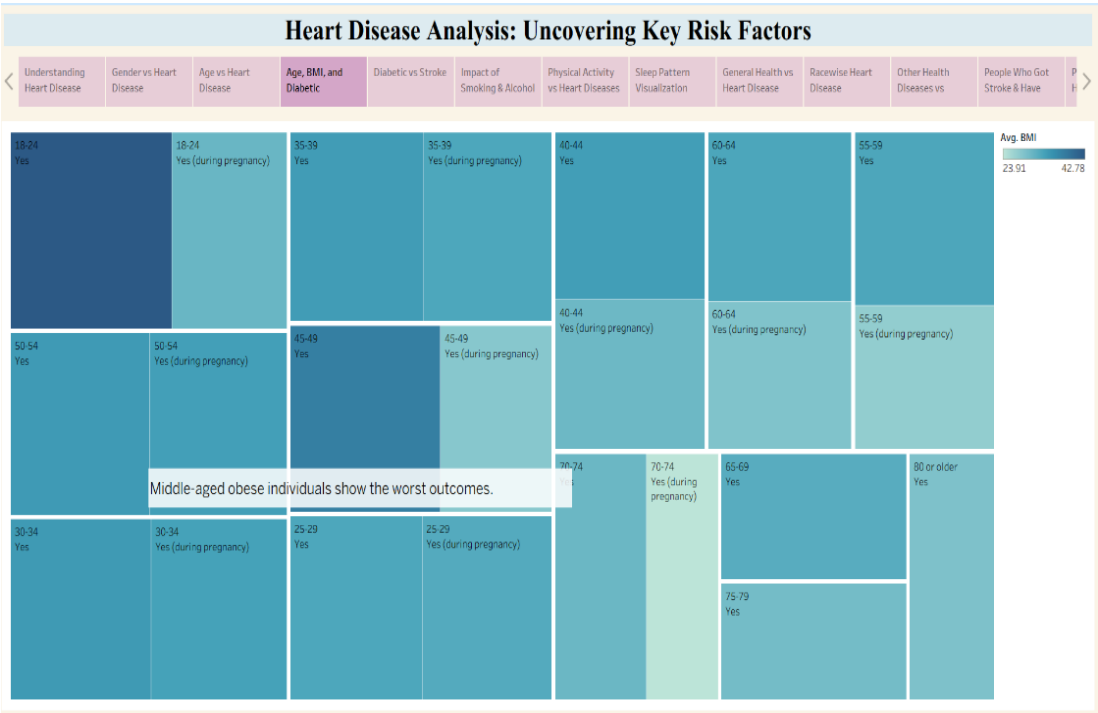
Include:

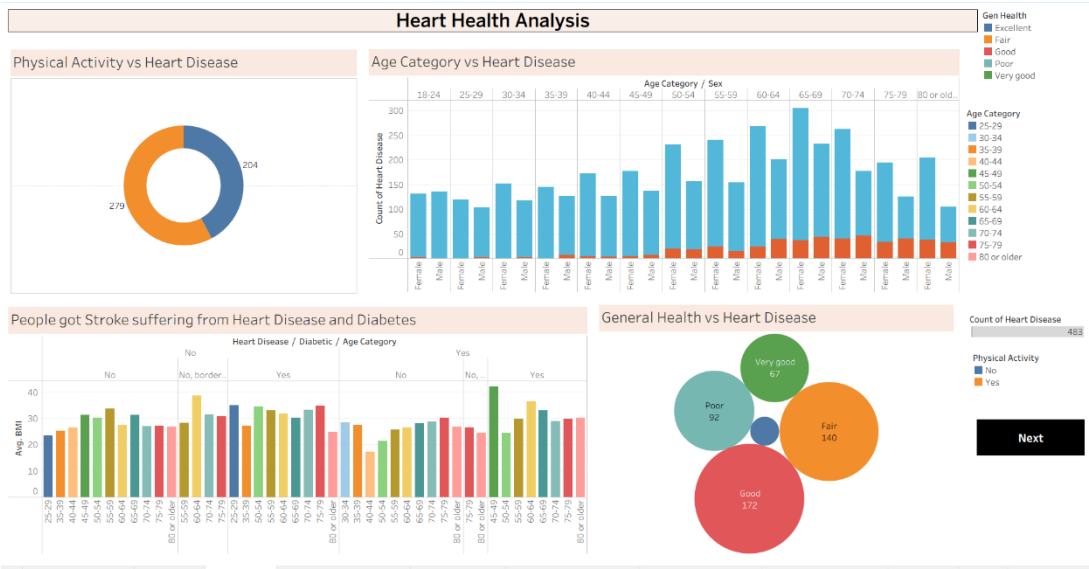
- Gender vs Heart Disease
- Age Category Analysis
- BMI Visualization

- Diabetes, Stroke, Kidney Disease charts
- Lifestyle dashboards
- Predictor Summary
- Final Story

Some screenshots:







8. ADVANTAGES & DISADVANTAGES

Advantages

- Easy to understand heart disease patterns
- Interactive dashboards
- Helps prioritize high-risk groups
- Supports preventive healthcare decisions

Disadvantages

- Depends on dataset quality
- Tableau Public have limitations
- Not a predictive model (only descriptive)

9. CONCLUSION

This project successfully identifies the major predictors of heart disease through powerful visual analytics. Interactive dashboards reveal actionable patterns, helping healthcare professionals understand risk and focus on prevention. The project demonstrates how data storytelling enhances awareness and supports informed decisions.

10. FUTURE SCOPE

- Build a predictive ML model (Logistic Regression, Random Forest)
- Add real-time health monitoring integration
- Create personalized patient dashboards
- Incorporate hospital datasets for deeper insights

11. APPENDIX

Dataset Link: https://drive.google.com/file/d/190Qmq27LeZZ_nWricP3Obl7ys_5otEsp/view

GitHub: https://github.com/Aadya133/Heart-Disease-Analysis

Project Demo Link:
https://drive.google.com/file/d/112UDvi9zD40FURfFcP_q3_4w6vXIcKNN/view?usp=sharing